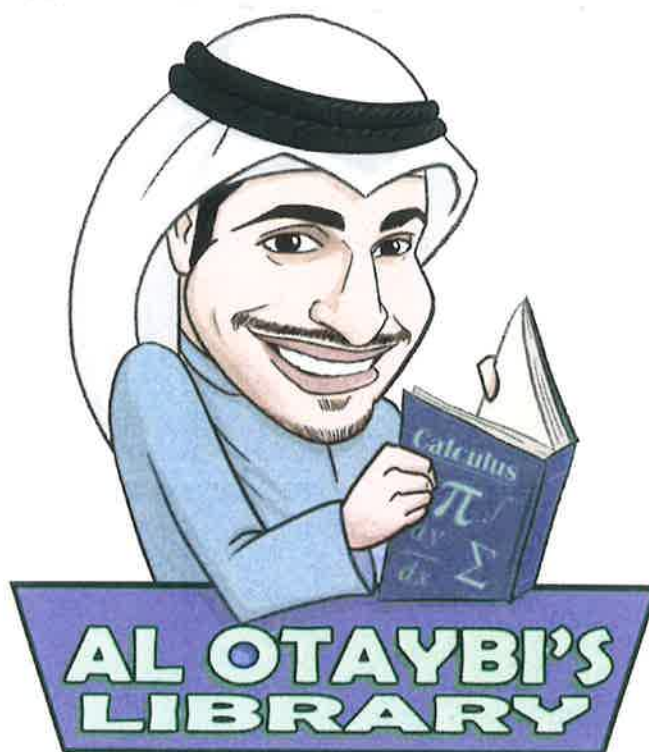


Kuwait University

2021/2022

Fall course



Calculus A

First

Section : 1.5 (part a)

1.5 Inverse Functions and Logarithms

(3)

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1.5 Inverse Functions and Logarithms

One – to – one function

A function f is called a **one – to – one** function if it **never** takes on the same value twice .

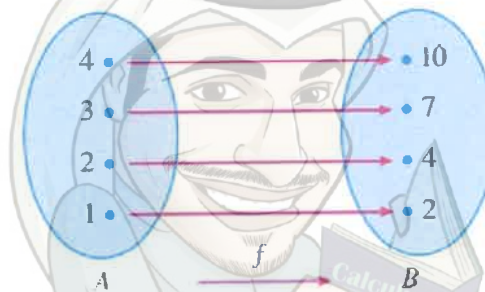
$$f(x_1) \neq f(x_2) \quad \text{whenever} \quad x_1 \neq x_2$$



Text book questions



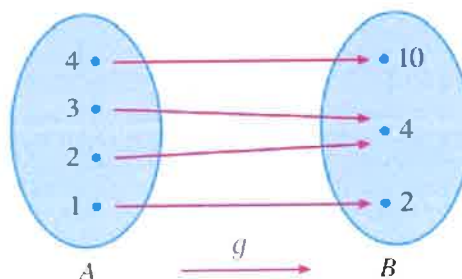
Determine whether the function f is one – to – one function or not ? (justify) .



$$\therefore f(1) \neq f(2) \neq f(3) \neq f(4)$$

$\therefore f$ is one-to-one.

Determine whether the function g is one – to – one function or not ? (justify) .



$$\therefore f(2) = f(3) = 4$$

$\therefore f$ is not one-to-one



Text book questions



A function f is given by a table of values.

Determine whether it is one – to – one .

x	1	2	3	4	5	6
$f(x)$	1.5	2.0	3.6	5.3	2.8	2.0

$$\therefore f(2) = f(6) = 2$$

$\therefore f$ is not one-to-one

A function f is given by a table of values.

Determine whether it is one – to – one .

x	1	2	3	4	5	6
$f(x)$	1.0	1.9	2.8	3.5	3.1	2.9

$$\therefore f(1) \neq f(2) \neq f(3) \neq f(4) \neq f(5) \neq f(6)$$

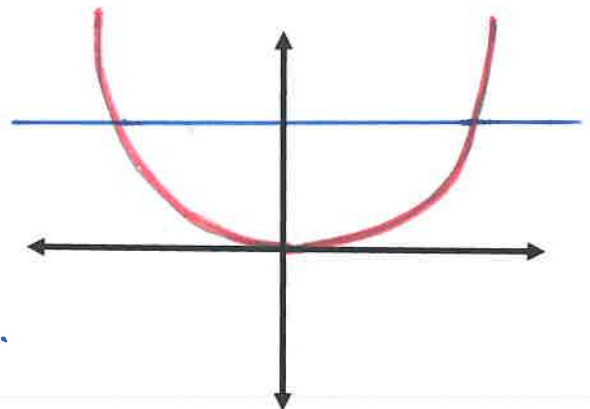
$\therefore f$ is one-to-one .

Determine whether f is one – to – one function or not ? (Justify) . where

$$f(x) = x^2$$

$\therefore$ There is horizontal line intersect the graph more than one time

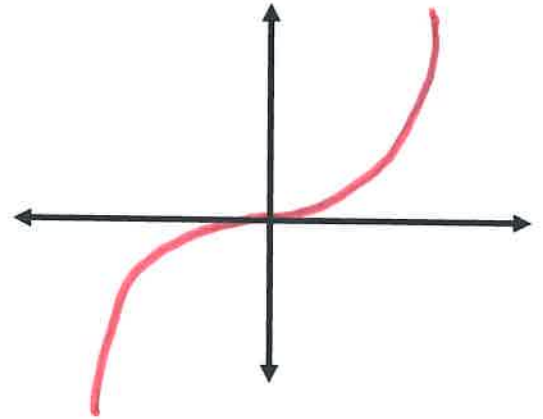
$\Rightarrow \therefore f$ is not one-to-one .



Determine whether f is one – to – one function or not ? (Justify) .

where $f(x) = x^3$

$\therefore$ No horizontal line intersect the graph more than one time
 $\Rightarrow \therefore f$ is one-to-one.



Text book questions

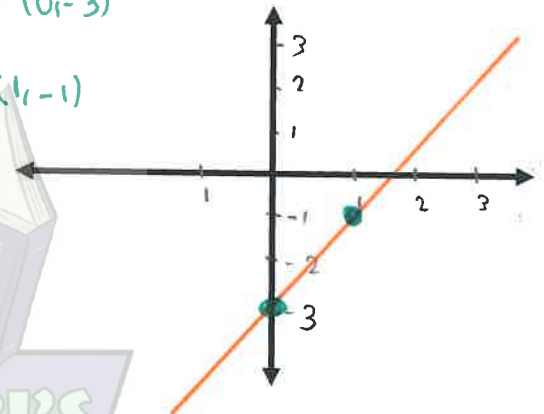


Determine whether f is one – to – one function or not ? (Justify) .

where $f(x) = 2x - 3$

$y = 2x - 3$
at $x=0 \rightarrow y=-3 \rightarrow (0, -3)$
at $x=1 \rightarrow y=-1 \rightarrow (1, -1)$

$\therefore$ No horizontal line intersect the graph more than one time
 $\Rightarrow \therefore f$ is one-to-one.



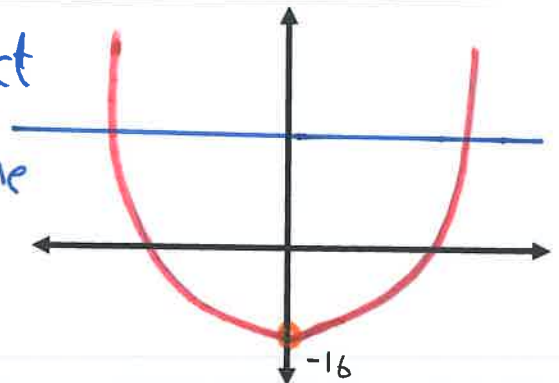
Text book questions



Determine whether f is one – to – one function or not ? (Justify) .

where $f(x) = x^4 - 16$

$\therefore$ There is horizontal line intersect the graph more than one time
 $\Rightarrow \therefore f$ is not one-to-one

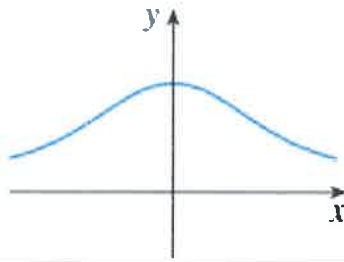




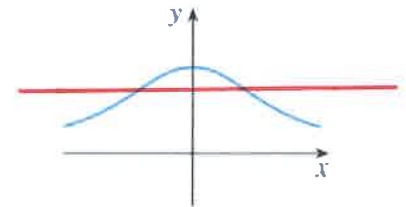
Text book questions



A function is given by a graph . Determine whether it is one-to-one.



$\therefore$ There is horizontal line intersect the graph more than one time



$\Rightarrow \therefore$ the function is NOT one-to-one



Text book questions



A function is given by a graph . Determine whether it is one-to-one.



$\therefore$ No horizontal line intersect the function more than one time .

$\Rightarrow \therefore$ the function is one-to-one.



Tricky questions



$$\text{Let } f(x) = \frac{2x+3}{x+3}$$

Determine whether f is one-to-one function or not

if $f(x_1) = f(x_2)$

$$\frac{2x_1+3}{x_1+3} = \frac{2x_2+3}{x_2+3}$$

$$(2x_1+3)(x_2+3) = (2x_2+3)(x_1+3)$$

$$\cancel{2x_1x_2} + 6x_1 + 3x_2 + 9 = \cancel{2x_2x_1} + 6x_2 + 3x_1 + 9$$

$$6x_1 - 3x_1 = 6x_2 - 3x_2$$

$$\div 3 \quad 3x_1 = 3x_2$$

$$\therefore \boxed{x_1 = x_2}$$

$\therefore f$ is one-to-one.

$$\text{Let } f(x) = \frac{x-1}{3x-1}$$

Determine whether f is one-to-one function or not

if $f(x_1) = f(x_2)$

$$\frac{x_1-1}{3x_1-1} = \frac{x_2-1}{3x_2-1}$$

$$(x_1-1)(3x_2-1) = (x_2-1)(3x_1-1)$$

$$\cancel{3x_1x_2} - x_1 - 3x_2 + 1 = \cancel{3x_1x_2} - x_2 - 3x_1 + 1$$

$$-x_1 + 3x_1 = -x_2 + 3x_2$$

$$\div 2 \quad 2x_1 = 2x_2$$

$$\boxed{x_1 = x_2}$$

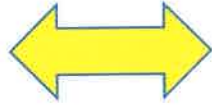
$\therefore f$ is one-to-one.



سوالف !!

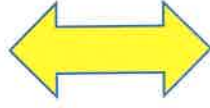


$$f(x) = 10x$$



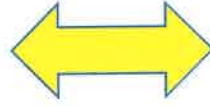
$$f(x) = \frac{1}{10}x$$

$$f(1) = 10(1) = 10$$



$$f(10) = \frac{1}{10} 10 = 1$$

$$f(2) = 10(2) = 20$$



$$f(20) = \frac{1}{10} 20 = 2$$

$$f(3) = 10(3) = 30$$



$$f(30) = \frac{1}{10} 30 = 3$$

$$f(a) = b$$



$$f(b) = a$$

Inverse function

هي الدالة التي تعطي نتائج عكسية لدالتها الأصلية



خواص أي دالة والانفرس لها



1) دومين يصير رينج



2) رينج يصير دومين

3) نقطة تصير صورة



4) صورة تصير نقطة

5) متى ما الصقت الدالة والانفرس لها ببعض كنسلوا بعض

If the Domain of $f(x) = \mathbb{R}$, Range of $f(x) = [0, \infty)$,
 $f(5) = 10$, and $f(-7) = 8$.

Find each of the following :

1) Domain of $f^{-1}(x)$ 2) Range of $f^{-1}(x)$

3) $f^{-1}(10)$ 4) $f^{-1}(8)$

5) $f(f^{-1}(5))$ 6) $f^{-1}(f(3))$

1) $D_{f^{-1}} = R_f = [0, \infty)$

2) $R_{f^{-1}} = D_f = \mathbb{R}$

3) $f^{-1}(10) = 5$

4) $f^{-1}(8) = -7$

5) ~~$f(f^{-1}(5)) = 5$~~

6) ~~$f^{-1}(f(3)) = 3$~~



Text book questions



Assuming that f is one-to-one function

a) If $f(4) = 7$, what is $f^{-1}(7)$?

b) If $f^{-1}(8) = 2$, what is $f(2)$?

a) $f^{-1}(7) = 4$

b) $f(2) = 8$



Text book question



If $f(x) = x^5 + x^3 + x$, find $f^{-1}(3)$ and $f(f^{-1}(2))$

$$f(0) = 0 + 0 + 0 = 0$$

$$f(1) = 1 + 1 + 1 = 3$$

$$\because f(1) = 3 \longrightarrow \therefore f^{-1}(3) = 1$$

$$f(f^{-1}(2)) = 2$$



Text book question



If $g(x) = 3 + x + e^x$, find $g^{-1}(4)$

$$g(0) = 3 + 0 + e^0 = 3 + 1 = 4$$

$$\because g(0) = 4 \longrightarrow \therefore g^{-1}(4) = 0$$

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LIBRARY

If $f(x) = 2x + \ln x$, find $f^{-1}(2)$

$$f(1) = 2(1) + \ln 1 = 2$$

$$\because f(1) = 2 \longrightarrow \therefore f^{-1}(2) = 1$$



Good to know

$$\ln(1) = 0$$

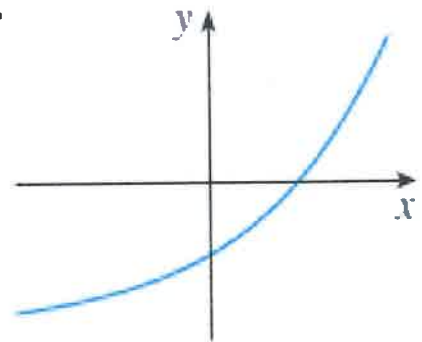


Text book question



If $f(x)$ is given by the following graph .

Is $f^{-1}(x)$ exist ??



- ∴ No horizontal line intersect the graph more than one time
- ∴ f is one-to-one
- ∴ f^{-1} is exist



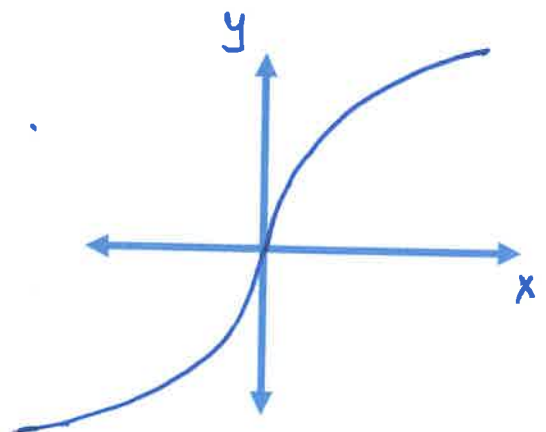
Text book question



If $g(x) = \sqrt[3]{x}$. Is $g^{-1}(x)$ exist ??

∴ No horizontal line intersect the graph more than one time .

- ∴ g is one-to-one
- ∴ g^{-1} is exist





خطوات إيجاد دالة الانفرس



1) $y = \dots$

2) أخلي x بطرف والباقي بطرف

3) أبدل $y \rightarrow x$
 $x \rightarrow f^{-1}(x)$



$$f^{-1}(x) = \dots$$



انتبه



إذا كانت الدالة مكونة من

بسط

مقام

وطلب الانفرس



For check



$$f(f^{-1}(x)) = x$$

$$f^{-1}(f(x)) = x$$



Text book question



Find the inverse function of $f(x) = x^3 + 2$

$$y = x^3 + 2$$

$$y - 2 = x^3$$

$$\therefore x = \sqrt[3]{y - 2}$$

$$\Rightarrow \therefore f^{-1}(x) = \sqrt[3]{x - 2}$$



Text book question



Find a formula for the inverse of the function

$$f(x) = 1 + \sqrt{2 + 3x}$$

$$y = 1 + \sqrt{2 + 3x}$$

$$y - 1 = \sqrt{2 + 3x}$$

$$2 + 3x = (y - 1)^2$$

$$\div 3 \quad 3x = (y - 1)^2 - 2$$

$$x = \frac{(y - 1)^2 - 2}{3}$$

$$\Rightarrow \therefore f^{-1}(x) = \frac{(x - 1)^2 - 2}{3}$$



Text book question



Let $f(x) = \sqrt{1-x^2}$, $0 \leq x \leq 1$. Find $f^{-1}(x)$

$$y = \sqrt{1-x^2}$$

$$y^2 = 1 - x^2$$

$$x^2 = 1 - y^2$$

$$x = \pm \sqrt{1-y^2}$$

$$\Rightarrow \therefore f^{-1}(x) = \sqrt{1-x^2}$$



Text book question



Find a formula for the inverse of the function

$$f(x) = \frac{4x-1}{2x+3}$$

$$y = \frac{4x-1}{2x+3}$$

$\times (2x+3)$

$$y(2x+3) = 4x-1$$

$$2xy + 3y = 4x - 1$$

$$2xy - 4x = -1 - 3y$$

$$\div [2y-4] \quad x[2y-4] = -1-3y$$

$$x = \frac{-1-3y}{2y-4} \Rightarrow \therefore f^{-1}(x) = \frac{-1-3x}{2x-4}$$

$$\text{Let } h(t) = \frac{t+1}{2t+1}$$

a) Find the domain of $h(t)$

b) Find the inverse function of $h(t)$

$$\begin{aligned} \text{a) } 2t+1 &= 0 \longrightarrow 2t = -1 \longrightarrow t = -\frac{1}{2} \\ \Rightarrow \therefore D_h &= \mathbb{R} \setminus \left\{ -\frac{1}{2} \right\} \end{aligned}$$

$$\text{b) } y = \frac{t+1}{2t+1}$$

$$y(2t+1) = t+1$$

$$2ty + y = t+1$$

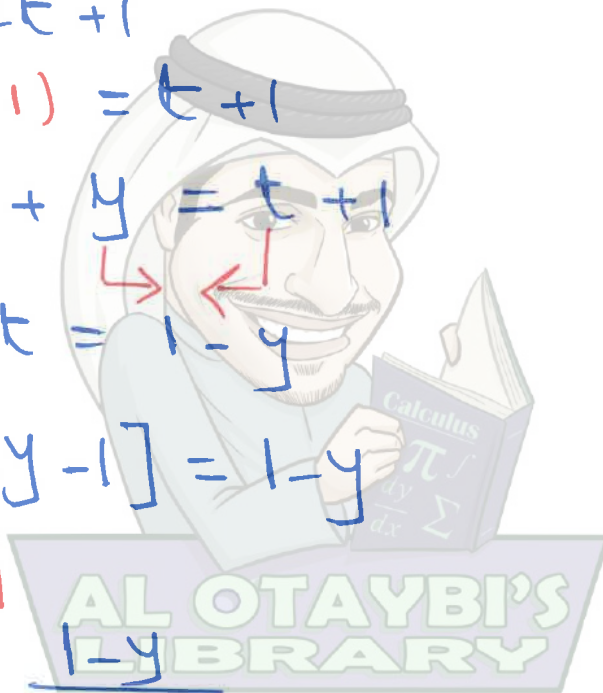
$$2ty - t = 1 - y$$

$$t[2y-1] = 1-y$$

$$\div [2y-1]$$

$$t = \frac{1-y}{2y-1}$$

$$h^{-1}(t) = \frac{1-t}{2t-1}$$





Tricky question



$$\text{Let } h(x) = x^2 - x - 2, \quad x \leq 0$$

Find $h^{-1}(x)$

$$y = x^2 - x - 2$$



$$x^2 - x - 2 - y = 0$$

$$a=1 \quad b=-1 \quad c=-2-y$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{1 \pm \sqrt{1 - 4(1)(-2-y)}}{2(1)}$$

$$= \frac{1 \pm \sqrt{1 + 8 + 4y}}{2}$$

$$= \frac{1 \pm \sqrt{9 + 4y}}{2}$$

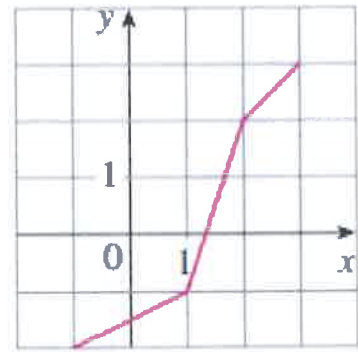
$$\because x \leq 0$$

$$\therefore x = \frac{1 - \sqrt{9 + 4y}}{2}$$

$$\Rightarrow \therefore h^{-1}(x) = \frac{1 - \sqrt{9 + 4x}}{2}$$

The graph of f is given

What is the value of $f^{-1}(-1)$?



$$f^{-1}(-1) = -2$$



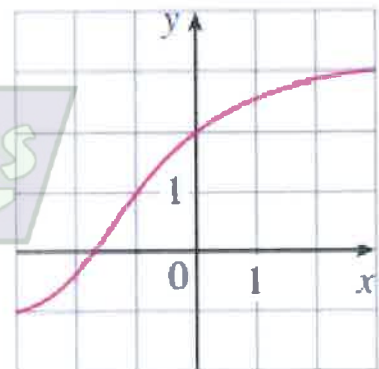
Text book questions



The graph of f is given

a) Why is f one-to-one?

b) What is the value of $f^{-1}(2)$?



a) $\because$ No horizontal line intersects the graph more than one time.

$\therefore f$ is 1-1

$$b) f^{-1}(2) = 1$$

Let $f(x) = \sqrt[5]{x+1}$ and $f^{-1}(x) = x^5 - 1$.

Show that $f(x)$ and $f^{-1}(x)$ are inverses to each other.

$$f(f^{-1}(x)) = \sqrt[5]{x^5 - 1 + 1}$$

$$= \sqrt[5]{x^5}$$

$$= x$$

$\therefore f(x)$ and $f^{-1}(x)$ are inverses to each other.





سوالف !!



Natural logarithmic function

$\ln x$

Laws of Logarithms

$$\ln(xy) = \ln x + \ln y$$

$$\ln\left(\frac{x}{y}\right) = \ln x - \ln y$$

$$\ln(x^y) = y \ln x$$



Zoom

$$\ln x^5 = 5 \ln x$$

$$\ln(\sin x)^8 = 8 \ln \sin x$$

$$\ln(\sin^8 x) = 8 \ln \sin x$$

$$\ln \sin x^8 = \text{لا تنطبق عليه الناحية}$$

$$\ln(x^2 + 1) = \text{لا تنطبق عليه الناحية}$$



سوالف !!



Logarithmic function

$$\log_a x$$

Laws of Logarithms

$$\log_a(xy) = \log_a x + \log_a y$$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$\log_a(x^y) = y \log_a x$$



القانون الذهبي



$$\log_a x = \frac{1}{\ln a} \ln x$$



Zoom

$$\log_3 x = \frac{1}{\ln 3} \ln x$$

$$\log_x 3 = \frac{1}{\ln x} \ln 3$$

$$\log_e x = \frac{1}{\ln e} \ln x = \ln x$$



مهم جدا



$\ln x$

inverse

e^x



Lets play !!



$$\ln e^x = x$$

$$\ln e^{x^2} = x^2$$

$$\ln e^{\sin x} = \sin x$$

$$e^{\ln x} = x$$

$$e^{\ln x^2} = x^2$$

$$e^{\ln \sin x} = \sin x$$



انتبه



$$\ln(e^x + 1) =$$

لا يمكن السطرب

$$\ln e^{x+1} = x+1$$



مهم جدا



$\log_a x$

inverse

a^x



Lets play !!



$$\log_3 3^x = x$$

$$\log_3 3^{x^2} = x^2$$

$$\log_3 3^{\sin x} = \sin x$$

$$4^{\log_4 x} = x$$

$$5^{\log_5 x^2} = x^2$$

$$2^{\log_2 \sin x} = \sin x$$



انتبه



$$4^{\log_7 x} =$$

لا يمكن الشطب
الأساسات مختلفة !



3 Text book questions



Find the exact value of the expression

$$\ln\left(\frac{1}{e^2}\right)$$

Solution # 1

$$= \ln e^{-2} = -2$$

Solution # 2

$$\begin{aligned} & \ln 1 - \ln e^2 \\ &= 0 - 2 = -2 \end{aligned}$$

Find the exact value of the expression

$$e^{\ln(\ln e^3)}$$

$$= e^{\ln(\ln e^3)} = e^{\ln 3} = 3$$

Find the exact value of the expression

$$e^{-\ln 2}$$

$$= e^{\ln 2^{-1}} = 2^{-1} = \frac{1}{2}$$

Find the exact value of the expression

$$e^{-2\ln 5}$$

$$= e^{\ln 5^{-2}} = 5^{-2} = \frac{1}{5^2} = \frac{1}{25}$$

Find the exact value of the expression

$$e^{2\ln 3}$$

$$= e^{\ln 3^2} = 3^2 = 9$$

Find the exact value of the expression

$$\log_{10} \sqrt{10}$$

$$= \cancel{\log_{10}} 10^{\frac{1}{2}} = \frac{1}{2}$$

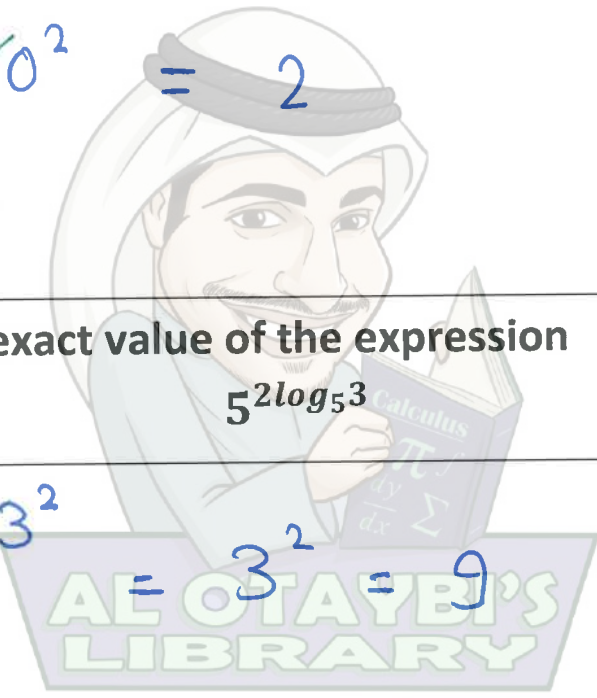
Find the exact value of the expression

$$\log_{10} 100$$

$$= \cancel{\log_{10}} 10^2 = 2$$

Find the exact value of the expression

$$5^{2\log_5 3}$$

$$= \cancel{5^{\log_5}} 3^2 = 3^2 = 9$$


Find the exact value of the expression

$$3^{-3\log_3 2}$$

$$= 3^{\log_3} 2^{-3} = 2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$



Text book questions



Find the exact value of the expression

$$\log_2 32$$

$$= \log_2 2^5 = 5$$

Find the exact value of the expression

$$\log_5 \frac{1}{125}$$

$$= \log_5 \frac{1}{5^3}$$
$$= \log_5 5^{-3} = -3$$

Find the exact value of the expression

$$\log_{10} 40 + \log_{10} 2.5$$

$$= \log_{10} (40 (2.5))$$

$$= \log_{10} (40 \cdot \frac{5}{2}) = \log_{10} 100$$

$$= \log_{10} 10^2 = 2$$



Text book questions



Express the given quantity as a single logarithm

$$\ln 20 - 2 \ln 2$$

$$= \ln 20 - \ln 2^2$$

$$= \ln 20 - \ln 4$$

$$= \ln\left(\frac{20}{4}\right) = \ln 5$$

Express the given quantity as a single logarithm

$$\ln b + 2 \ln c - 3 \ln d$$

$$= \ln b + \ln c^2 - \ln d^3$$

$$= \ln \left[\frac{b \cdot c^2}{d^3} \right]$$

Express the given quantity as a single logarithm

$$\frac{1}{3} \ln(x+2)^3 + \frac{1}{2} \left[\ln x - \ln(x^2 + 3x + 2)^2 \right]$$

$$= \frac{1}{3} \ln(x+2)^3 + \frac{1}{2} \ln x - \frac{1}{2} \ln(x^2 + 3x + 2)^2$$

$$= \ln \left[(x+2)^{\cancel{3}^{\frac{1}{3}}} \right] + \ln x^{\frac{1}{2}} - \ln \left[(x^2 + 3x + 2)^{\cancel{2}^{\frac{1}{2}}} \right]$$

$$= \ln(x+2) + \ln x^{\frac{1}{2}} - \ln(x^2 + 3x + 2)$$

$$= \ln \frac{(x+2) x^{\frac{1}{2}}}{x^2 + 3x + 2}$$

$$= \ln \frac{(x+2) \sqrt{x}}{(x+1)(x+2)} = \ln \frac{\sqrt{x}}{x+1}$$

Find the exact value of the expression

$$\log_{10} 4 + 2\log_{10} 5$$

July 2017

$$\log_{10} 4 + 2\log_{10} 5$$

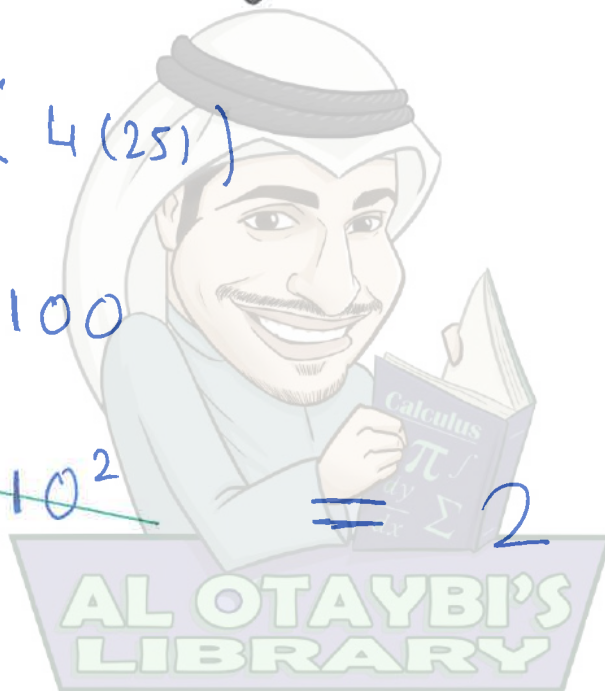
$$\log_{10} 4 + \log_{10} 5^2$$

$$\log_{10} 4 + \log_{10} 25$$

$$\log_{10} (4(25))$$

$$\log_{10} 100$$

$$\log_{10} 10^2 = 2$$



Find the exact value of the expression

$$\log_2 6 - \log_2 15 + \log_2 20$$

$$\log_2 \frac{6 \cdot 20}{15}$$

$$\log_2 \frac{2 \cdot 6 \cdot (4)}{3}$$

$$\log_2 2(4)$$

$$\log_2 8 = \log_2 2^3 = 3$$

Find the exact value of the expression

$$\log_3 100 - \log_3 18 - \log_3 50$$

$$\log_3 \frac{100}{18 \cdot 50}$$

$$\log_3 \frac{2}{18}$$

$$\log_3 \frac{1}{9} = \log_3 \frac{1}{3^2} = \log_3 3^{-2} = -2$$



Do NOT forget !!



القانون الذهبي



$$\log_a x = \frac{1}{\ln a} \ln x$$



Text book questions



Find the exact value of each expression

$$\log_8 2$$

$$= \frac{1}{\ln 8} \cdot \ln 2 = \frac{1}{\ln 2^3} \ln 2$$

$$= \frac{1}{3 \ln 2} \ln 2 = \frac{1}{3}$$

Find the exact value of each expression

$$\log_8 60 - \log_8 3 - \log_8 5$$

$$= \log_8 \frac{60}{3(5)} = \log_8 \frac{20}{5}$$

$$= \log_8 4$$

$$= \frac{1}{\ln 8} \ln 4 = \frac{1}{\ln 2^3} \ln 2^2$$

$$= \frac{1}{3 \ln 2} \cdot 2 \ln 2 = \frac{2}{3}$$

Find the domain of

$$f(x) = \ln(e^x - 5)$$

$$\begin{aligned} e^x - 5 &> 0 \\ e^x &> 5 \\ \cancel{\ln e^x} &> \ln 5 \\ x &> \ln 5 \\ D_f &= (\ln 5, \infty) \end{aligned}$$

Find the domain of

$$f(x) = \ln(2^{x+1} - 6)$$

$$\begin{aligned} 2^{x+1} - 6 &> 0 \\ 2^{x+1} &> 6 \\ \cancel{\log_2 2^{x+1}} &> \log_2 6 \\ x+1 &> \log_2 6 \\ x &> \log_2 6 - 1 \\ D_f &= (\log_2 6 - 1, \infty) \end{aligned}$$

Find the domain of

$$f(x) = \sqrt{2e^x - 2}$$

$$\begin{aligned} 2e^x - 2 &\geq 0 \\ 2e^x &\geq 2 \\ \div 2 & \\ e^x &\geq 1 \\ \cancel{\ln e^x} &\geq \ln 1 \\ x &\geq 0 \rightarrow D_f = [0, \infty) \end{aligned}$$

Find the domain of

$$f(x) = \frac{1}{e^x - 3}$$

$$\begin{aligned} e^x - 3 &\neq 0 \\ e^x &\neq 3 \\ \cancel{\ln e^x} &\neq \ln 3 \\ x &\neq \ln 3 \\ D_f &= \mathbb{R} \setminus \{\ln 3\} \end{aligned}$$

Find the domain of

$$f(x) = \ln(e^x + 5)$$

$$\begin{aligned} e^x + 5 &> 0 \\ e^x &> -5 \\ D_f &= \mathbb{R} \end{aligned}$$

Find the domain of

$$f(x) = \frac{e^x}{e^x + 3}$$

$$\begin{aligned} e^x + 3 &\neq 0 \\ e^x &\neq -3 \\ D_f &= \mathbb{R} \end{aligned}$$



Text book questions



Solve each equation for x

$$e^{7-4x} = 6$$

Domain = $\mathbb{R}$

$$\cancel{e^{7-4x}} = \ln 6$$

$$7 - 4x = \ln 6$$

$$7 - \ln 6 = 4x$$

$$\therefore x = \frac{7 - \ln 6}{4}$$

Solve each equation for x

$$\ln(3x - 10) = 2$$

Domain : $3x - 10 \geq 0$
 $3x \geq 10$
 $x \geq \frac{10}{3}$

$\Rightarrow \therefore \text{Domain} = (\frac{10}{3}, \infty)$

$$\cancel{e^{\ln(3x-10)}} = e^2$$

$$3x - 10 = e^2$$

$$3x = e^2 + 10$$

$$\therefore x = \frac{e^2 + 10}{3}$$

Solve each equation for x

$$\ln(x^2 - 1) = 3$$

Domain : $x^2 - 1 \geq 0$
 $x^2 \geq 1$

$\Rightarrow \therefore \text{Domain} : \mathbb{R} \setminus [-1, 1]$

$$\cancel{e^{\ln(x^2-1)}} = e^3$$

$$x^2 - 1 = e^3$$

$$x^2 = e^3 + 1$$

$$x = \pm \sqrt{e^3 + 1}$$

Solve each equation for x

$$2^{x-5} = 3$$

Domain : $\mathbb{R}$

$$\cancel{\log_2 2^{x-5}} = \log_2 3$$

$$x - 5 = \log_2 3$$

$$x = \log_2 3 + 5$$

Solve the equation $\ln(x^2 - 2x - 2) = 0$

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$$\ln(x^2 - 2x - 2) = 0$$

$$e^{\ln(x^2 - 2x - 2)} = e^0$$

$$x^2 - 2x - 2 = 1$$



$$x^2 - 2x - 2 - 1 = 0$$

$$x^2 - 2x - 3 = 0$$

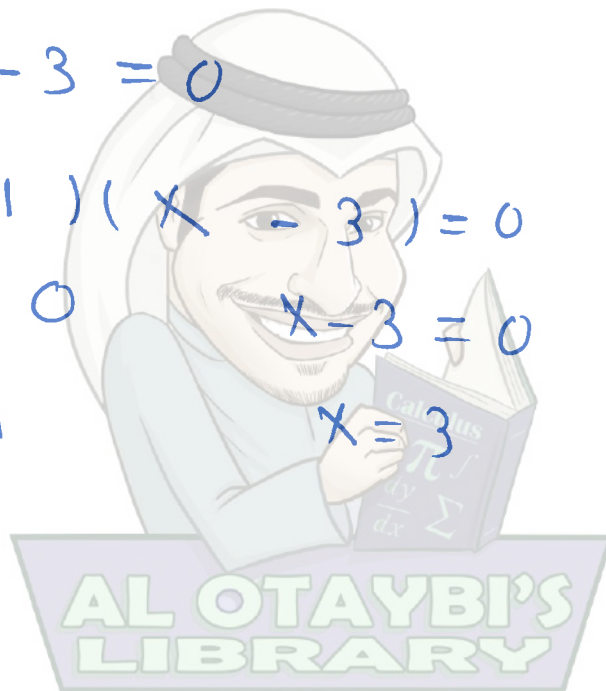
$$(x + 1)(x - 3) = 0$$

$$x + 1 = 0$$

$$x = -1$$

$$x - 3 = 0$$

$$x = 3$$



Solve for x

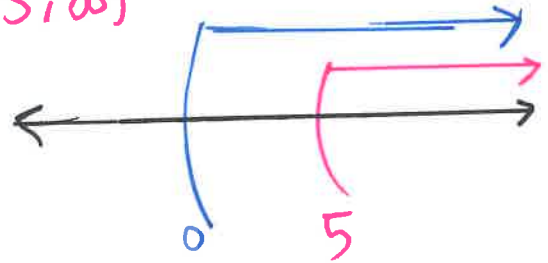
$$\log_6 x + \log_6 (x - 5) = 1$$

Domain :

$$x \geq 0 \\ (0, \infty)$$

$$x - 5 \geq 0 \\ x \geq 5 \\ (5, \infty)$$

$$\Rightarrow \therefore \text{Domain} = (5, \infty)$$



$$\log_6 [x(x-5)] = 1$$

$$6^{\log_6 [x(x-5)]} = 6^1$$

$$x(x-5) = 6$$

$$x^2 - 5x = 6$$

$$x^2 - 5x - 6 = 0$$

$$(x+1)(x-6) = 0$$

$$x+1 = 0$$

$$x = -1$$

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$$x-6 = 0$$

$$x = 6$$

$$\Rightarrow \therefore x = 6$$

Text book questions

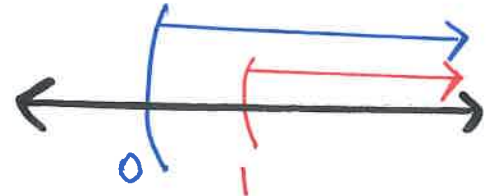
Solve each equation for x

$$\ln(\ln x) = 1$$

Domain: $x > 0$
 $(0, \infty)$

$\ln x > 0$
 ~~$e^{\ln x} > e^0$~~
 $x > 1$
 $(1, \infty)$

$\cap$



$\Rightarrow$ Domain = $(1, \infty)$

$$\ln(\ln x) = 1$$

~~$e^{\ln(\ln x)} = e^1$~~

$$\ln x = e$$

~~$e^{\ln x} = e^e$~~

$$x = e^e$$

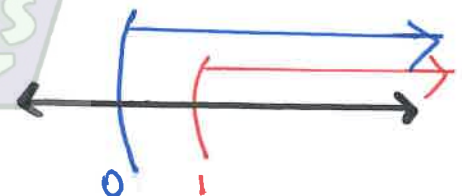
Solve each equation for x

$$\ln x + \ln(x-1) = 1$$

Domain: $x > 0$
 $(0, \infty)$

$x-1 > 0$
 $x > 1$
 $(1, \infty)$

$\cap$



$\Rightarrow$ Domain = $(1, \infty)$

$$\ln x + \ln(x-1) = 1$$

$$\ln [x(x-1)] = 1$$

~~$e^{\ln [x(x-1)]} = e^1$~~

$$x(x-1) = e$$

$$x^2 - x = e$$

$$x^2 - x - e = 0$$

$$a=1 \quad b=-1 \quad c=-e$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{1 \pm \sqrt{1 - 4(1)(-e)}}{2(1)} = \frac{1 \pm \sqrt{1+4e}}{2}$$

$\Rightarrow \therefore x = \frac{1 + \sqrt{1+4e}}{2} \in (1, \infty)$



Text book questions



Solve each equation for x

$$e^{2x} - 3e^x + 2 = 0$$

Domain = $\mathbb{R}$

$$(e^x - 1)(e^x - 2) = 0$$

$$e^x - 1 = 0$$

$$e^x = 1$$

~~$$\ln e^x = \ln 1$$~~

$$x = 0$$

$$e^x - 2 = 0$$

$$e^x = 2$$

~~$$\ln e^x = \ln 2$$~~

$$x = \ln 2$$

$$x^2 - 3x + 2 = 0$$

$$(x - 1)(x - 2) = 0$$



Tricky question



Solve each equation for x

$$e^{2x} - 5e^x - 6 = 0$$

Domain = $\mathbb{R}$

$$(e^x + 1)(e^x - 6) = 0$$

$$e^x + 1 = 0$$

$$e^x = -1$$

None

$$e^x - 6 = 0$$

$$e^x = 6$$

~~$$\ln e^x = \ln 6$$~~

$$x = \ln 6$$

$$x^2 - 5x - 6 = 0$$

$$(x + 1)(x - 6) = 0$$

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